

Title: Beyond Two-Point Statistics: A New Framework for Integrating Traditional and Alternative Clustering Methods

Abstract:

On the largest scales, the matter distribution of the Universe is well described by a Gaussian random field. This property has made two-point statistics the primary tools for extracting cosmological information from large-scale structure. However, gravitational collapse generates highly nonlinear, non-Gaussian structure on smaller scales, and because these probes are difficult to model and analyze, much of the nonlinear regime is often discarded in current analyses. We propose to develop a rigorous framework that identifies and combines the most informative measurements across a broad range of statistics to integrate Gaussian and non-Gaussian information from cosmic structure. Our strategy selects probes from traditional two-point statistics and alternative clustering methods that capture higher-order information and integrates them into a single, optimized framework using a greedy algorithm. The proposed strategy is critical as cosmology enters the era of precision surveys, where experiments such as the Dark Energy Spectroscopic Instrument, Euclid space telescope mission, and the Nancy Grace Roman Space Telescope are producing datasets so large and precise that analyses can no longer afford to discard substantial fractions of the available data. Unlocking the information encoded in nonlinear structure strengthens tests of the Λ CDM framework and provides insight into galaxy formation, the galaxy–halo connection, and the physical processes governing cosmic structure formation.

- **Scientific Justification**

Recovering Cosmological Information from Nonlinear Structure in Precision Surveys

On the largest scales, the matter distribution of the Universe is well described by a Gaussian random field (Peebles 1980). This is a well-established result, supported by observations of the cosmic microwave background and large-scale galaxy surveys, which consistently show that primordial density fluctuations follow near-perfect Gaussian statistics. As a result, two-point statistics have become the primary tools for extracting cosmological information from large-scale structure, as they fully characterize a Gaussian field. The dominance of two-point statistics has driven significant theoretical and observational progress, forming the backbone of modern cosmological inference.

However, gravitational collapse drives the formation of highly nonlinear and non-Gaussian structures on small scales, where a significant fraction of cosmological information resides (Scoccimarro 2000), yet these regimes remain difficult to model and are often excluded from analyses (Bernardeau et al. 2002). With next-generation surveys such as the Dark Energy Spectroscopic Instrument (DESI) delivering data of unprecedented precision, the inability to incorporate nonlinear information is becoming a dominant limitation, weakening cosmological constraints and restricting our ability to test fundamental physics.

We propose a statistical framework that identifies and combines the most informative measurements across multiple probes, integrating Gaussian and non-Gaussian information from large-scale structure into a single optimized pipeline. We target the DESI spectroscopic galaxy survey because its large volume, high number density, and precise redshift measurements make it uniquely suited to test and validate new methods in regimes where nonlinear information is typically discarded. Our strategy is to construct an optimized joint statistic by selecting complementary two-point and higher-order clustering observables and combining them using a greedy algorithm, validated on simulations before application to DESI data. **This framework will enable robust extraction of information from nonlinear structure, strengthening constraints on Λ CDM and improving our understanding of galaxy formation and the galaxy–halo connection.** More broadly, it will provide a transferable methodology for future large-scale structure surveys, ensuring they can fully exploit the information content of the nonlinear Universe.

Nonlinear mode coupling causes information loss in traditional large-scale structure analyses

Nonlinear structure formation induces mode coupling, transferring information from the power spectrum into higher-order correlations and generating non-Gaussian features in the density field (Scoccimarro 2000; Bernardeau et al. 2002). As a result, the highest signal-to-noise modes in modern surveys lie precisely in the nonlinear regime where our practical statistical description is least complete (Desjacques et al. 2016). However, these nonlinear scales are often excluded from standard analyses, meaning that a significant fraction of

cosmological information is discarded, typically on scales below $\sim 20 h^{-1}$ Mpc, limiting the constraining power of large-scale structure measurements (Senatore and Zaldarriaga 2015). Incorporating non-Gaussian structure in clustering analyses would enable the recovery of information lost to mode coupling, improving parameter constraints beyond two-point statistics (Cheng and Ménard 2021). Such clustering analyses involve joint use of Gaussian and non-Gaussian observables, which is increasingly critical as surveys like DESI approach sample-variance limits on large scales.

There is currently no systematic framework that can quantify the marginal information gain of different observables in the presence of mode coupling, nor reliably determine how to combine them into an optimal joint statistic. Our work will develop a method to evaluate the information content of correlated large-scale structure observables and construct efficient, non-redundant combinations of Gaussian and non-Gaussian statistics. Using a greedy algorithm, we will iteratively select and combine complementary statistics to identify a near-optimal set of probes that efficiently captures both Gaussian and non-Gaussian information, enabling a unified and information-maximizing analysis of large-scale structure.

This proposal: Emulator-Based Framework Will Extract Linear & Nonlinear Information from DESI-like Data

Using summary statistics, including standard two-point statistics as well as higher-order and alternative clustering measures sensitive to non-Gaussian structure, we target the cosmological information encoded in both the linear and nonlinear regimes of large-scale structure. The DESI catalog is uniquely suited to this endeavor: its unprecedented combination of survey volume, redshift coverage, and galaxy number density yields high signal-to-noise measurements that extend deep into the nonlinear regime, where much of the statistical constraining power resides. As a result, DESI analyses are currently limited more by modeling frameworks than by the quality or abundance of the data, providing ideal conditions for jointly leveraging a wide range of linear and nonlinear probes.

We propose an emulator-based framework for modeling galaxy clustering using DESI-like mock catalogs, as schematically illustrated in Figure 1, which shows how linear and nonlinear information is combined within a unified pipeline. The mocks will be generated using AbacusSummit dark matter simulations with AbacusHOD (Maksimova et al. 2021; Yuan et al. 2022) to reproduce DESI survey characteristics and to measure both two-point and non-Gaussian clustering observables. These measurements will train an emulator that maps cosmological and nuisance parameters to observables, while a greedy algorithm will select subsets of statistics that maximize information gain. The resulting optimized data vector will be used for cosmological parameter inference, and the full pipeline will be applied to real DESI data once validated.

This work will deliver a validated simulation-based inference pipeline that can be directly deployed within DESI analyses to combine complementary clustering probes across linear and nonlinear regimes. This product directly addresses the loss of small-scale information in traditional two-point-only analyses, enabling more complete and robust cosmological constraints from DESI data.

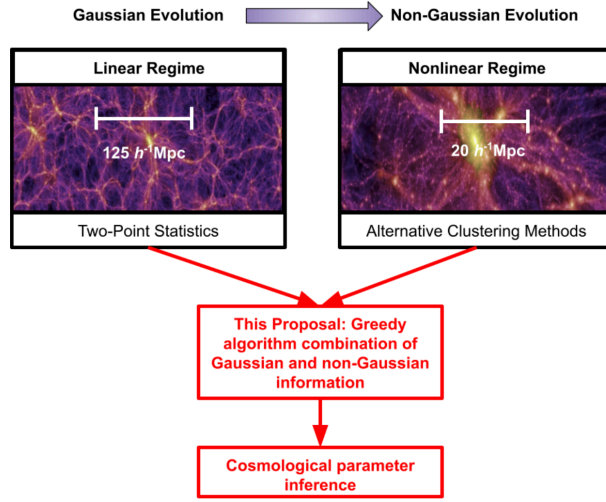


Figure 1: **By combining information from both the linear and non-linear regimes of large-scale structure, this program will improve cosmological constraints relative to analyses that use either regime alone.** The figure contrasts the linear (Gaussian) large-scale regime with the nonlinear (non-Gaussian) small-scale regime and highlights the different statistics used to probe each. The left panel shows a large-

scale view of the density field in the linear regime, while the right panel zooms into smaller scales where nonlinear gravitational collapse produces clusters, filaments, and voids (Springel et al. 2005). The indicated $\sim 20 h^{-1}\text{Mpc}$ scale marks the approximate transition where nonlinear effects become important. Standard analyses rely primarily on linear statistics and therefore fail to capture the cosmological information redistributed into higher-order correlations and non-Gaussian features by nonlinear gravitational evolution. However, naively combining linear and nonlinear statistics produces large, highly covariant data vectors that require careful modeling. We propose a principled algorithm to identify and combine the most informative components of linear and nonlinear statistics into a single optimized data vector. This approach enables efficient recovery of non-Gaussian information while preserving the robustness of linear probes, yielding significantly stronger cosmological constraints than are possible with linear statistics alone.

Gaussian and Non-Gaussian Information Extraction Will Maximize the Scientific Return of Precision Surveys

Our proposed program addresses a central limitation in large-scale structure cosmology: the inability to fully exploit information from nonlinear clustering. By enabling the joint use of Gaussian and non-Gaussian probes within galaxy clustering analyses, our work will advance the field toward a more complete picture of the Universe’s structure and evolution by incorporating small-scale modes that are currently discarded in analyses, without requiring new observations.

The proposed work is critical to undertake now as cosmology has entered a precision era in which large surveys are no longer limited by data volume but by analysis frameworks, and further gains in constraining models of the Universe hinge on accessing the nonlinear regime. By targeting information in small-scale structure, our work will directly align with DESI’s core science goals of constraining dark energy, testing ΛCDM , and refining measurements of structure growth through large-scale structure analyses. This work will deliver a publicly available emulator and optimization framework applicable to next-generation surveys such as Euclid and the Nancy Grace Roman Space Telescope, maximizing their scientific return while providing a more unified view across cosmology, galaxy formation, and high-dimensional inference methods.

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